

James McAllister

github.com/Jpmac110205 · linkedin.com/in/jpmac1102 · jpmac1102@outlook.com · (267) 768-8243

TECHNICAL SUMMARY

Software Engineering student experienced in software development, AI/ML, distributed systems, and API design, with hands-on experience building production AI systems, automation pipelines, and cloud-based applications. Proficient in Python, Java, JavaScript/TypeScript, and SQL, with experience across RAG, REST APIs, authentication, CI/CD, and full-stack development.

EDUCATION

Bachelor of Science in Software Engineering (ABET Accredited)

May 2028

Shippensburg University of Pennsylvania; Wood Honors College

Minors: Computer Science, Mathematics, Data Science

WORK EXPERIENCE

IT Innovation Intern | NJM Insurance Group

June — August 2026

- Designed and implemented an advanced RAG pipeline utilizing recursive chunking, query rewriting, and reranking to achieve 96% retrieval accuracy across an enterprise knowledge base.
- Engineered an automated evaluation and testing framework measuring system accuracy, error rate, latency, and throughput across a production pipeline, achieving 90% end-to-end accuracy at 370ms average latency.
- Engineered an automated document synchronization system processing 760+ enterprise documents via REST APIs, webhooks, and GitHub Actions, implementing incremental/full-sync modes and deduplication logic to eliminate redundant uploads at scale.
- Collaborated with solution architects to assess AI proofs of concept, presenting technical feasibility and business adoption strategies for enterprise deployment.

PROJECTS

Prodigy | Solo AI Engineer

December 2025 — April 2026

- Designed and deployed a full-stack AI application using React, FastAPI, and PostgreSQL, integrating Google Calendar/Tasks APIs for real-time productivity workflows. <https://prodigyaiassistant.onrender.com/>
- Built a RAG pipeline with document parsing, variable chunking, embedding generation, and semantic retrieval for Q&A.
- Engineered OAuth2 authentication and authorization with automatic token refresh and session management across 20+ REST API endpoints.
- Implemented a multi-layer memory architecture managing session state, persistent knowledge base, and conversational context.

LifeLens | Lead Designer and Engineer

August — December 2025

- Developed a full-stack deep learning computer vision system using PyTorch, CNNs, and transfer learning with ResNet, achieving 90% validation accuracy across 20,000+ images and implementing Grad-CAM model explainability.
- Architected a RESTful FastAPI backend to serve ML models and process multimodal data, integrating structured medical data with image classification results and explainable risk assessments.

BeneFit | Founder and Lead Software Engineer

June — September 2025

- Engineered a full-stack cross-platform mobile application using the Flutter framework for iOS and Android.
- Designed and implemented a scalable NoSQL database architecture using Firebase Firestore with real-time synchronization and Firebase Authentication for secure user management.
- Integrated USDA FoodData API enabling real-time search across 300,000+ food items with macro and calorie tracking.
- Developed a gamification engine with XP-based leveling, 13+ challenges, an achievement system, step tracking features, and social features (friend streaks, rankings), utilizing feedback from a more than 10-person beta testing group to drive engagement.

CaesarOS (In Progress) | AI Architect & Engineer

August 2026 — Present

- Architecting a multi-agent AI orchestration platform using LangGraph to coordinate specialized agents through shared workflow state.
- Integrating Discord, Google Workspace APIs, ChromaDB, and Firebase to build an AI ecosystem for productivity, knowledge retrieval, and automation.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, Dart, C, SQL, NoSQL

Frameworks & libraries: React, Flutter, FastAPI, PyTorch, NumPy, ChromaDB, LangChain, LangGraph

Cloud & development: Git, GitHub, GitHub Actions, Firebase, REST APIs, OpenAI API, Google Cloud

AI & machine learning: Retrieval-Augmented Generation, LLMs, AI Agents, Vector Databases, Deep Learning, CNNs, Transfer Learning